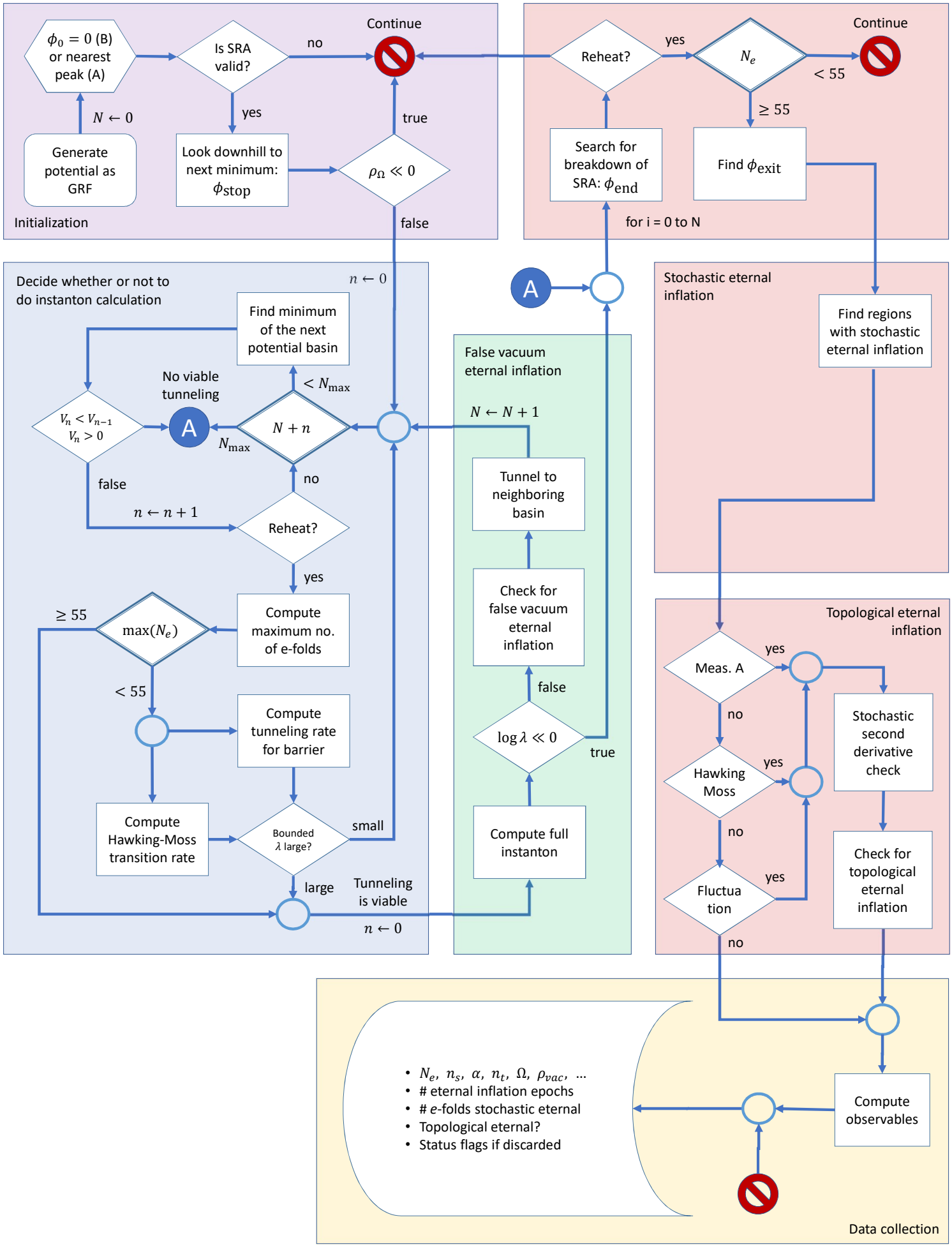
Initialization
$\phi_0 = 0$ (B) or nearest peak (A)
Is SRA valid?
no
Continue
yes
$N \leftarrow 0$
Generate potential as GRF
Look downhill to next minimum: ϕ_{stop}
$\rho_\Omega \ll 0$
true
false
for i = 0 to N
Reheat?
yes
N_e
Continue
< 55
≥ 55
Search for breakdown of SRA: ϕ_{end}
Find ϕ_{exit}
Decide whether or not to do instanton calculation
$n \leftarrow 0$
Find minimum of the next potential basin
No viable tunneling
$< N_{\max}$
$V_n < V_{n-1}$
$V_n > 0$
A
$N + n$
$N_{\max}$
false
no
$n \leftarrow n + 1$
Reheat?
yes
≥ 55
$\max(N_e)$
Compute maximum no. of e-folds
< 55
Compute tunneling rate for barrier
Compute Hawking-Moss transition rate
Bounded λ large?
small
large
Tunneling is viable
$n \leftarrow 0$
A
False vacuum eternal inflation
$N \leftarrow N + 1$
Tunnel to neighboring basin
Check for false vacuum eternal inflation
false
$\log \lambda \ll 0$
true
Compute full instanton
Stochastic eternal inflation
Find regions with stochastic eternal inflation
Topological eternal inflation
Meas. A
yes
no
Hawking Moss
yes
no
Fluctua tion
yes
no
Stochastic second derivative check
Check for topological eternal inflation
Compute observables
N_e, n_s, α, n_t, Ω, ρ_{vac}, ...
eternal inflation epochs
e-folds stochastic eternal
Topological eternal?
Status flags if discarded
Data collection